

Assignment 2

Topic: MFCC Feature Extraction and Comparative Analysis of Indian Languages (Question 2).

Date: 03-04-2025

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Github link: repository

1 Objective

This assignment involves two main tasks focused on audio signal processing and machine learning for language classification:

Task A: Feature Extraction and Analysis

The first task involved downloading the Audio Dataset with 10 Indian Languages from Kaggle and implementing a Python program to extract Mel-Frequency Cepstral Coefficients (MFCCs)[1] from each audio sample. We generated and visualized MFCC spectrograms for representative samples from three selected languages (Hindi, Marathi, and Punjabi) to analyze their acoustic characteristics. By comparing these spectrograms, we identified both distinct patterns and similarities in their spectral representations, which formed the basis for our language classification approach. The visualization revealed language-specific signatures in the MFCCs that could potentially be used for automated identification.

Task B: Language Classification

Building upon the extracted MFCC features, we developed a classifier system to predict the language of audio samples. After evaluating several machine learning approaches, we selected a neural network architecture for its ability to capture complex patterns in the MFCC data. The implementation included essential preprocessing steps such as feature normalization and proper train-test splits to ensure robust model evaluation. Throughout the development process, we maintained modular code organization and comprehensive documentation to facilitate reproducibility and future enhancements to the system. The classifier's performance was subsequently evaluated using standard metrics to assess its effectiveness in distinguishing between the target languages.

2 Task A: Feature Extraction and Analysis

Mel-Frequency Cepstral Coefficients (MFCCs) are a widely used feature representation in speech processing that capture the short-term power spectrum of sound through several key steps. First, they emulate human auditory perception by applying a Mel-scale transformation, which mimics the nonlinear frequency resolution of the human ear. Next, they extract crucial spectral envelope details using the Discrete Cosine Transform (DCT), effectively compressing the spectral information into a compact set of coefficients. Importantly, MFCCs discard pitch-related information to focus specifically on timbral and textural features that are most relevant for speech characterization. In language recognition tasks, these coefficients prove particularly valuable as they help distinguish unique phonetic and prosodic patterns that differentiate languages. However, their effectiveness can be limited by sensitivity to noise and speaker-dependent variations, which may introduce variability in the extracted features.[1]

2.1 MFCC Spectrogram Analysis

Figures 1, 2, and 3 show MFCC spectrograms for Hindi, Marathi, and Punjabi samples, respectively.

Table 1 summarizes the mean MFCC coefficients (first two dimensions) for Marathi, Hindi, and Punjabi samples, highlighting inter-language variability.

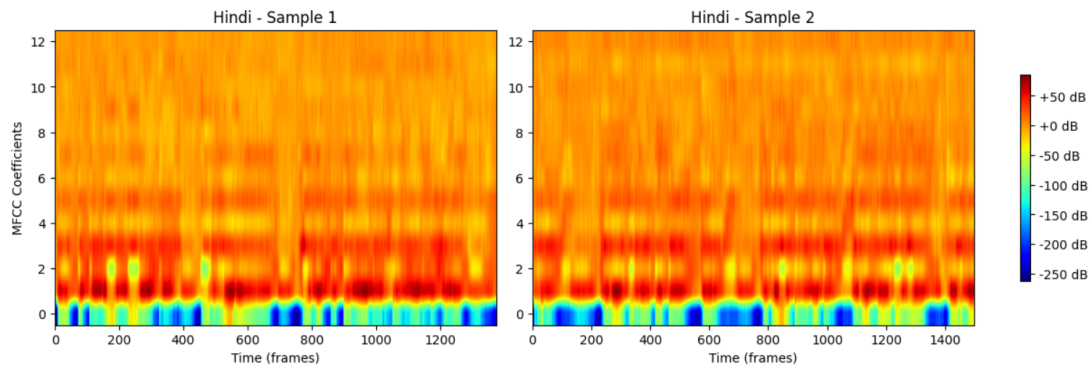


Figure 1: MFCC spectrogram for Hindi).

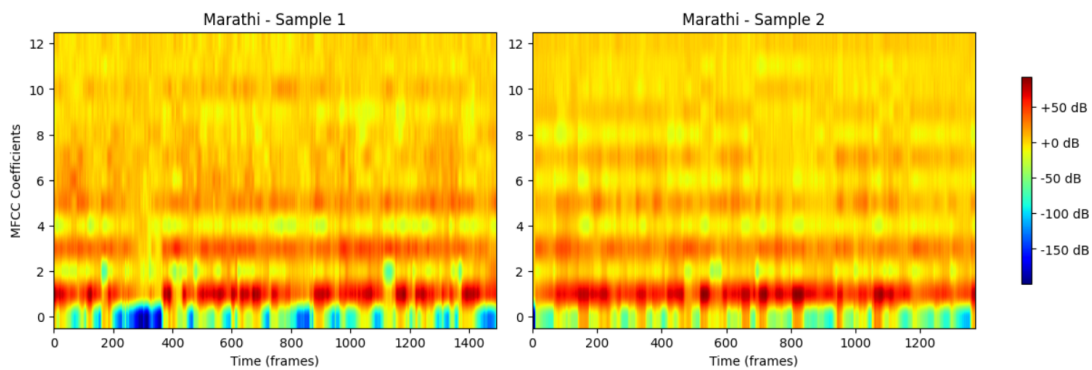


Figure 2: MFCC spectrogram for Marathi.

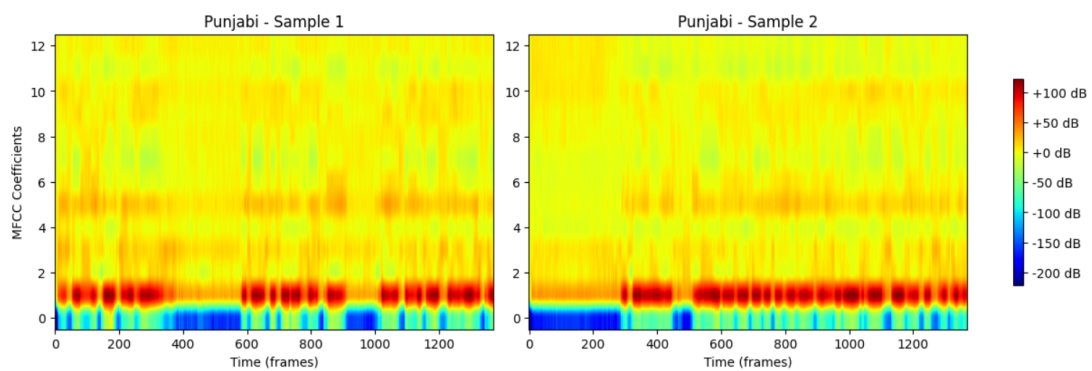


Figure 3: MFCC spectrogram for Punjabi.

Key Insights

The statistical analysis reveals distinct acoustic characteristics across languages. Marathi samples exhibit positive MFCC means (ranging from 0.86 to 3.73), suggesting dominant mid-frequency energy in their spectral patterns. In contrast, Hindi shows consistently negative means (between -3.72 and -2.53), indicating stronger low-frequency components. Punjabi occupies an intermediate position with near-zero means (-0.97 to -0.76), reflecting a balanced spectral distribution. Notably, intra-language consistency is observed: samples from the same language maintain similar coefficient magnitudes, exemplified by Hindi's narrow range (~ 1.2 difference between samples). This uniformity within languages supports MFCC's utility for capturing language-specific features, though the absolute differences highlight measurable acoustic boundaries between linguistic groups.

Table 1: Mean MFCC Coefficients (First Two Dimensions) by Language

Language (Sample)	MFCC Mean (1st Coef.)	MFCC Mean (2nd Coef.)
Marathi - Sample 1	2.048464	0.8596929
Marathi - Sample 2	3.7191103	3.727614
Hindi - Sample 1	-2.5293324	-2.5301905
Hindi - Sample 2	-3.7225397	-3.7236917
Punjabi - Sample 1	-0.7639154	-0.7639872
Punjabi - Sample 2	-0.9742848	-0.97230405

2.2 Limitations of MFCCs

MFCCs are not foolproof due to:

- **Speaker Variability:** Pitch and vocal tract differences alter coefficients across speakers.
- **Background Noise:** Additive noise distorts spectral estimates.
- **Regional Accents:** Dialects may overlap in MFCC space, reducing separability.

For robust classification, MFCCs should be combined with prosodic features (e.g., pitch, duration) or deep learning models.

3 Task B: Language Classification Task

We developed a custom dataloader[2] to manage the data processing pipeline efficiently. The audio files underwent preprocessing including truncation and padding to handle variable lengths, followed by MFCC feature extraction. The final dataset comprised 205,466 training samples and 51,367 test samples across 10 Indian languages.

3.1 Model Architecture and Training

We implemented a neural network classifier with three hidden layers for language identification, using MFCC coefficients as input features. The model was trained with cross-entropy loss function and the Adam optimizer, achieving an overall accuracy of 86.70% on unseen test data. As shown in Figure 4, the confusion matrix reveals specific challenges in distinguishing between Gujarati and Punjabi languages, despite strong performance across most other language classes. This architecture successfully captured the spectral patterns in MFCC features while maintaining computational efficiency, though the observed confusion between certain language pairs indicates areas for potential improvement in feature representation or model capacity.

3.2 Results and Analysis

Table 2 presents the detailed classification report. The model performed well for most languages (F1-score more than 0.95) except Gujarati (0.09) and Punjabi (0.65). This aligns with our MFCC analysis showing Punjabi's atypical low-pitched characteristics causing confusion with Gujarati. The limitations stem from

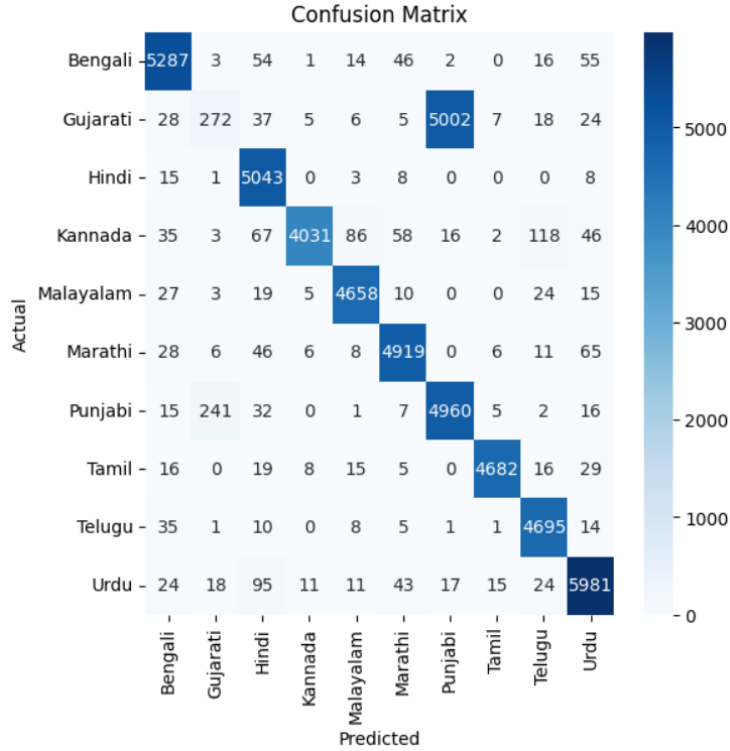


Figure 4: Confusion matrix showing classification performance across 10 languages

MFCC's inability to capture syntactic/semantic information. Some possible solutions to this problem include incorporating prosodic features (pitch, duration) alongside MFCCs, using attention mechanisms to model temporal dependencies or augmenting data with speaker-normalized samples

Table 2: Classification Report (Precision, Recall, F1-score)

Language	Precision	Recall	F1-score
Bengali	0.96	0.97	0.96
Gujarati	0.50	0.05	0.09
Hindi	0.93	0.99	0.96
Kannada	0.99	0.90	0.95
Malayalam	0.97	0.98	0.97
Marathi	0.96	0.97	0.96
Punjabi	0.50	0.94	0.65
Tamil	0.99	0.98	0.98
Telugu	0.95	0.98	0.97
Urdu	0.96	0.96	0.96

References

- [1] T. Bäckström, O. Räsänen, A. Zewoudie, P. Pérez Zarazaga, L. Koivusalo, S. Das, E. Gómez Mellado, M. Bouafif Mansali, D. Ramos, S. Kadiri, P. Alku, and M. H. Vali, *Introduction to Speech Processing*, 2nd ed., 2022. [Online]. Available: <https://speechprocessingbook.aalto.fi>. DOI: 10.5281/zenodo.6821775
- [2] A. Paszke et al., “PyTorch: An Imperative Style, High-Performance Deep Learning Library,” in *Advances in Neural Information Processing Systems 32* (NeurIPS 2019). [Online]. Available: <https://pytorch.org>